SMART POLICE PATROL ROBOT

A Low-Cost Autonomous Safety-Monitoring Prototype

Team Name	**Robotics & Things**	**Stage**	Stage 1 — Abstract Submission
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1. OBJECTIVE

The proposed Smart Police Patrol Robot is a compact autonomous surveillance platform designed for campuses, warehouses, and restricted zones where continuous human patrol is difficult. It aims to reduce routine patrol effort while offering obstacle-aware movement, remote visual monitoring, and real-time local warning alerts.

2. METHODOLOGY & ARCHITECTURE

An Arduino Uno serves as the central processing unit, controlling a 4WD motor platform via an L298N driver. Navigation decisions rely on an HC-SR04 ultrasonic sensor for distance measurement and obstacle avoidance. A servo-mounted ESP32-CAM module provides wireless live video transmission with pan/tilt scanning. Local visual and audible warnings are triggered via red/blue LEDs and a buzzer upon detecting obstacles.

3. SYSTEM OVERVIEW & FLOW

POWER & SENSING		CONTROL & LOGIC		ACTUATION & OUTPUT
Rechargeable Battery (Power Regulation) **HC-SR04 Sensor** (Obstacle Distance)	→	**Arduino Uno** (Main Controller / Navigation) **Servo Motor** (Directional Scanning)	→	**L298N Driver** (4WD Geared DC Motors) **ESP32-CAM** (Wi-Fi Live Video Stream)
Working Sequence: Patrol → Detect Obstacle → Stop → Scan Directions → Select Clear Path → Resume Movement				

4. PROPOSED OUTCOME

The prototype successfully demonstrates autonomous patrol, obstacle avoidance, remote video streaming, and immediate warning alerts. Designed modularly for scalability, it enables future additions like GPS tracking and AI video analytics. Intended beneficiaries include public-safety teams, campus security, and warehouse operators requiring cost-effective patrol solutions.